



2014
HSC Trial
EXAMINATION

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Student Number

Class (Please circle)

M1 M2

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved Calculators may be used
- A table of standard integrals is provided
- In Questions 11 – 16 show relevant mathematical reasoning and/or calculations

Total Marks – 100

Section I Questions 1 – 10 10 marks

Allow about 15 minutes for this section

Section II Questions 11 – 16 90 marks

Allow about 2 hour and 45 minutes for this section

SECTION I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

- 1 If $z = 1 + 2i$ and $w = 3 - i$,

$z - \overline{w}$ is?

(A) $3i - 2$

(B) $4 + 3i$

(C) $i - 2$

(D) $4 + i$

- 2 The directrices of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ are:

(A) $x = \pm \frac{16\sqrt{7}}{7}$

(B) $x = \pm \frac{\sqrt{7}}{16}$

(C) $x = \pm \frac{16}{5}$

(D) $x = \pm \frac{5}{16}$

3 The factorisation of $x^4 + 7x^2 - 18$ over the complex number field is:

(A) $(x - i\sqrt{2})(x + i\sqrt{2})(x - 3)(x + 3)$

(B) $(x - i\sqrt{2})(x - i\sqrt{2})(x + 3)(x + 3)$

(C) $(x - 3i)(x - 3i)(x + \sqrt{2})(x + \sqrt{2})$

(D) $(x - 3i)(x + 3i)(x - \sqrt{2})(x + \sqrt{2})$

4 If $f(x) = \frac{x(x-1)}{x-2}$, which of the following lines will be an asymptote $y = f(x)$?

(A) $y = x + 1$

(B) $y = x - 2$

(C) $y = x - 1$

(D) $y = 0$

5 If α, β, γ are the roots of $x^3 + x - 1 = 0$, then an equation with roots $\frac{\alpha}{2}, \frac{\beta}{2}, \frac{\gamma}{2}$ is:

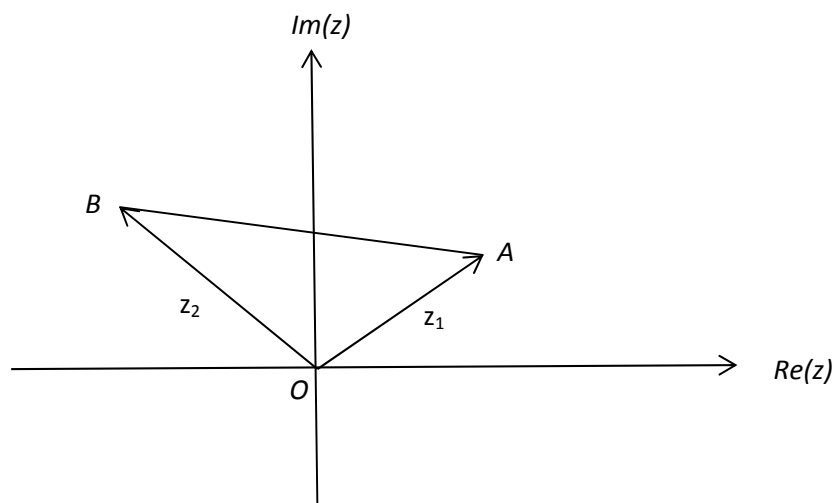
(A) $\frac{x^3}{8} + \frac{x}{2} - 1 = 0$

(B) $\frac{8}{x^3} + \frac{2}{x} - 1 = 0$

(C) $8x^3 + 2x - 1 = 0$

(D) $-8x^3 - 2x - 1 = 0$

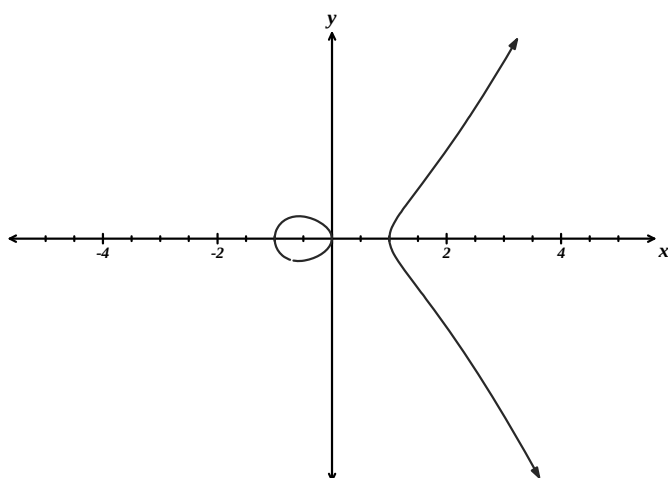
- 6 In the Argand diagram below, vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent the complex numbers z_1 and z_2 respectively where $|z_2| = 2|z_1|$ and $\angle AOB = \frac{2\pi}{3}$.



Vector $\overrightarrow{AB}$ represents?

- (A) $(-2 + i\sqrt{3})z_1$
- (B) $(2 - i\sqrt{3})z_1$
- (C) $(-\sqrt{3} + 2i)z_1$
- (D) $(\sqrt{3} - 2i)z_1$

- 7 The diagram shows the graph of $y^2 = f(x)$.



Which expression best represents the function $f(x)$

- (A) $x^2(x-1)$
- (B) $x^2(1-x)$
- (C) $x(x^2-1)$
- (D) $x(1-x^2)$
- 8 Four female and four male athletes are arranged in a row for the presentation of prizes. In how many ways can this be done if the males and females must alternate?

- (A) $4! \times 4!$
- (B) $2 \times 4! \times 4!$
- (C) $4! \times 5!$
- (D) $2 \times 4! \times 5!$

- 9** What is the natural domain of the function

$$f(x) = \frac{1}{2} \left(x\sqrt{x^2 - 1} - \log_e \left(x + \sqrt{x^2 - 1} \right) \right) ?$$

- (A) $x \leq -1$ or $x \geq 1$
- (B) $-1 \leq x \leq 1$
- (C) $x \geq 1$
- (D) $x \leq -1$

- 10** Four digit numbers are formed from the digits 1, 2, 3 and 4. Each digit is used once only. The sum of all the numbers that can be formed is?

- (A) 266,640
- (B) 66,660
- (C) 44,440
- (D) 6,666

SECTION II

90 marks

Attempt Questions 11-16

Allow about 2 hour and 45 minutes for this section

Answer each question in a SEPARATE Writing booklet. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks) Use a SEPARATE writing booklet.

- (a) (i) Write $\sqrt{3} + i$ in modulus/argument form. 2
- (ii) Hence, express $z = \frac{-1+i}{\sqrt{3}+i}$ in modulus/argument form and 3
- find the exact value of $\cos \frac{7\pi}{12}$ in simplest form.
- (b) (i) Find constants A, B, C such that $\frac{x^2 + 3x - 4}{x^2 - x - 2} \equiv A + \frac{Bx + C}{x^2 - x - 2}$ 2
- (ii) Hence find $\int \frac{x^2 + 3x - 4}{x^2 - x - 2} dx$ 2
- (c) Sketch the region on the Argand diagram where the inequalities 2
- $|z| \leq 4$ and $0 \leq \arg(z+2) \leq \frac{\pi}{4}$ both hold.
- (d) Using the substitution $x = 2 \sin \theta$, evaluate $\int_0^{\sqrt{3}} x^2 (4 - x^2)^{-\frac{5}{2}} dx$ 4

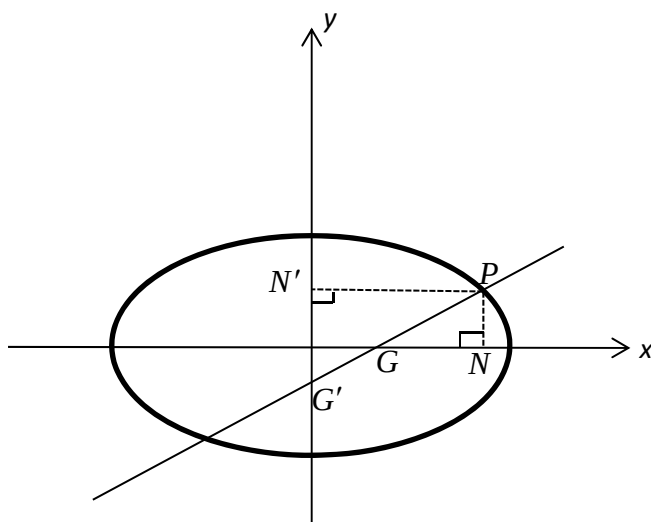
Question 12 (15 Marks) Use a SEPARATE writing booklet.

- (a) (i) Show that $2 + i$ is a root of $x^3 - 11x + 20 = 0$. 1
- (ii) Hence or otherwise solve $x^3 - 11x + 20 = 0$. 2

- (b) The diagram below shows the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse.

The normal to the ellipse at P meets the major and minor axes of the ellipse in G and G' respectively. N and N' are the feet of the perpendiculars from P to the major and minor axes respectively.

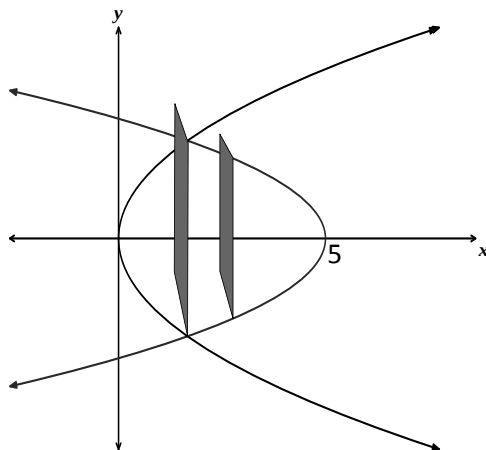


- (i) Show that the equation of the normal at P is $ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2$. 2
- (ii) Show that the ratio $OG : ON = e^2 : 1$ 2
- (iii) Find the ratio of the area of $\triangle PNG : \triangle PN'G'$. 1

Question 12 continues on page 9

Question 12 (continued)

- (c) Given the polynomial $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$ has a root of multiplicity 3, solve $P(x) = 0$. **3**
- (d) The base of a solid is the region bounded by the parabolas $x = y^2$ and $x = 3 - 2y^2$. Vertical cross-sections are squares perpendicular to the x -axis as shown in the diagram. **4**

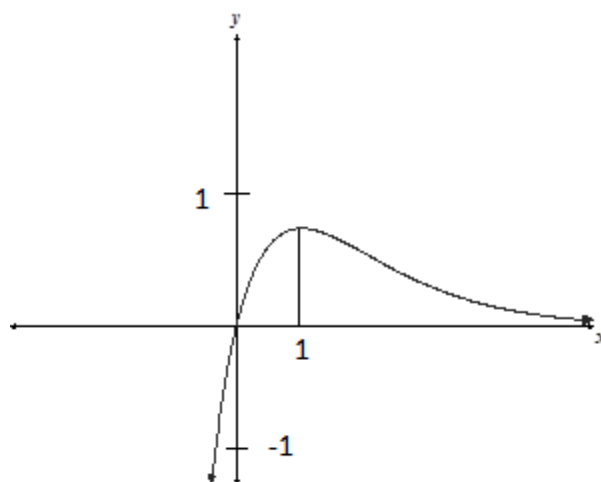


Find the volume of the solid.

End of Question 12

Question 13 (15 Marks) Use a SEPARATE writing booklet.

- (a) The diagram below shows the graph of $y = f(x)$.



NOT TO SCALE

As $x \rightarrow \infty$, $f(x) \rightarrow 0$. Draw large (one-third page), separate sketches of the following curves. Show any important features.

(i) $y = e^{f(x)}$ 2

(ii) $y = \sin^{-1} f(x)$ 3

- (b) From a group of eleven teachers, four will be chosen to form a committee. One teacher will be chairperson of the committee and the others will serve as members. 2

In how many ways can the committee be formed?

Question 13 continues on page 11

Question 13 (continued)

(c) $\int \frac{1}{\sqrt{3-(x^2-2x)}} dx$ 3

(d) $\int \tan^{-1} x \, dx$ 2

(e) Prove $\cos x - 1 + \frac{x^2}{2} > 0$, if $x \neq 0$. 3

End of Question 13

Question 14 (15 Marks) Use a SEPARATE writing booklet.

(a) Given $f(x) = \frac{3}{14}x^{2/3}\left(x^4 - \frac{7}{2}x^2 + 7\right)$

(i) Show that $y = f(x)$ is an even function. 1

(ii) The graph of $y = f(x)$ has two stationary points. 2

One stationary point is at $\left(1, \frac{27}{28}\right)$. Determine its nature.

(iii) Write down the coordinates of any point(s) where $f'(x)$ is undefined. 1

(iv) Sketch a graph of $y = f(x)$ showing all important features. 2

(b) (i) Using the substitution $t = \tan \frac{\theta}{2}$ show that: 3

$$I = \int_0^{\pi} \frac{d\theta}{a + b \cos \theta} = \lim_{h \rightarrow \infty} \int_0^h \frac{2dt}{(a-b)(t^2 + c^2)}$$

where $c^2 = \frac{a+b}{a-b}$ and $a > b > 0$.

(ii) Hence show that $I = \frac{\pi}{\sqrt{a^2 - b^2}}$ 2

(c) Using the method of cylindrical shells, find the volume of the solid of revolution formed when the area bounded by $y = \sin x$, the x -axis, between $x = 0$ and $x = \pi$, is rotated about the y -axis.

Question 15 (15 Marks) Use a SEPARATE writing booklet.

- (a) (i) Show that $(3 + \sqrt{5})^n + (3 - \sqrt{5})^n$ is divisible by 2^n 1

for $n = 1$ and $n = 2$.

- (ii) Use mathematical induction to prove that $(3 + \sqrt{5})^n + (3 - \sqrt{5})^n$ 3

is divisible by 2^n , for integer $n \geq 1$.

- (b) A non-real number w is such that $|w| = 1$. 3

If $z = \frac{1+w}{1-w}$ find the locus of z as w moves on the complex number plane.

- (c) A rubber ball of mass 7 kg, falls from rest, from the top of a building.

While falling the ball experiences a resistive force $\frac{7v^2}{10}$, where v is the velocity

of the ball. Take g , acceleration due to gravity, as $g = 10 \text{ ms}^{-2}$.

- (i) Show that $\ddot{x} = 10 - \frac{v^2}{10}$, where x is the distance the ball has fallen. 1

- (ii) Find the terminal velocity of the ball as it falls. 1

- (iii) Show that $v^2 = 100 \left(1 - e^{-\frac{x}{5}} \right)$ 3

- (iv) After hitting the ground the ball rises vertically such that 3

$\ddot{X} = -10 - \frac{V^2}{10}$, where V is the velocity of the ball as it rises and

X is the distance the ball rises.

Find the time that it takes for the ball to rise to its maximum height

if initially $V = \frac{10}{\sqrt{3}} \text{ m/s}$.

Question 16 (15 Marks) Use a SEPARATE writing booklet.

- (a) (i) Use de Moivre's Theorem to show that: 3

$$\tan 7\theta = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6} \text{ where } t = \tan \theta$$

- (ii) Hence show that the roots of the equation $x^3 - 21x^2 + 35x - 7 = 0$ 3

$$\text{are } \tan^2\left(\frac{\pi}{7}\right), \tan^2\left(\frac{2\pi}{7}\right) \text{ and } \tan^2\left(\frac{3\pi}{7}\right)$$

- (iii) Deduce that: 2

$$\sec^4\left(\frac{\pi}{7}\right) + \sec^4\left(\frac{2\pi}{7}\right) + \sec^4\left(\frac{3\pi}{7}\right) = 416$$

- (b) (i) Show that: 2

$$\frac{1}{1-x} = 1 + x + x^2 + \dots + x^{n-1} + \frac{x^n}{1-x}$$

- (ii) By letting $x = -t$, show that: 3

$$\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^{n-1} \frac{x^n}{n} + T_n(x)$$

$$\text{Where } T_n(x) = (-1)^n \int_0^x \frac{t^n}{1+t} dt.$$

- (iii) Show that $T_n(x) \rightarrow 0$ for $0 \leq x \leq 1$. 1

- (iv) Hence express $\log_e 2$ as a series writing the first 5 terms. 1

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Section 1 – 10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this part

Use the multiple-choice answer sheet overpage.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A 0 B ☒ C 0 D 0

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A 0 B ☒ C 0 D 0

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C 0 D 0

correct

↙



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Student Number

Class (Please circle)

12M1 12M2 12M3 12M4

12M5 12M6 11M1

Mathematics Extension 2

Multiple Choice Answer Sheet

Instructions: Colour in the circle next to the letter that represents the correct answer.

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

NBHS SOLUTIONS 2014 Extension 2 Trial HSC

Section 1

10 marks

Questions 1 – 10 (1 mark each)

Question 1 (1 mark)

Outcomes Assessed: E3

Targeted Performance Bands: E2

Solution	Answer	Mark
$z - \bar{w} = (1 + 2i) - (3 + i)$ $= -2 + i$	C	1

Question 2 (1 mark)

Outcomes Assessed: E3

Targeted Performance Bands: E2

Solution	Answer	Mark
$a = 4, b = 3$ $b^2 = a^2(1 - e^2)$ $9 = 16(1 - e^2)$ $e^2 = \frac{7}{16}$ $e = \frac{\sqrt{7}}{4}$ $x = \pm \frac{16\sqrt{7}}{7}$	A	1

Question 3 (1 mark)

Outcomes Assessed: E4

Targeted Performance Bands: E3

Solution	Answer	Mark
$x^4 + 7x^2 - 18$ $= (x^2 + 9)(x^2 - 2)$ $= (x^2 - (3i)^2)(x - \sqrt{2})(x + \sqrt{2})$ $= (x - 3i)(x + 3i)(x - \sqrt{2})(x + \sqrt{2})$	D	1

Question 4 (1 mark)**Outcomes Assessed:** E6**Targeted Performance Bands:** E3

Solution	Answer	Mark
$\frac{x(x-1)}{x-2} = \frac{x^2-x}{x-2}$ $= \frac{x^2-x-2+2}{x-2}$ $= \frac{(x-2)(x+1)+2}{x-2}$ $= x+1 + \frac{2}{x-2}$ $\therefore \text{asymptote is } y=x+1$	A	1

Question 5 (1 mark)**Outcomes Assessed:** E4**Targeted Performance Bands:** E3

Solution	Answer	Mark
Since α, β, γ satisfy $x^3 + x - 1 = 0$ $\frac{\alpha}{2}, \frac{\beta}{2}, \frac{\gamma}{2}$ satisfy $(2x)^3 + (2x) - 1 = 0$ $\therefore$ the required equation is $8x^3 + 2x - 1 = 0$	C	1

Question 6 (1 mark)**Outcomes Assessed:** E3**Targeted Performance Bands:** E3

Solution	Answer	Mark
Since $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ $= 2\text{cis}\left(\frac{2\pi}{3}\right)z_1 - z_1$ $= \left[2\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) - 1\right]z_1$ $= (-2 + i\sqrt{3})z_1$	A	1

Question 7 (1 mark)**Outcomes Assessed: E6****Targeted Performance Bands: E2**

Solution	Answer	Mark
Zeros at $x = -1, 1$ and 0 as $x \rightarrow \infty, y \rightarrow \infty$	C	1

Question 8 (1 mark)**Outcomes Assessed: E5****Targeted Performance Bands: E2**

Solution	Answer	Mark
With female first, e.g.: $F_1 M_1 F_2 M_2 F_3 M_3 F_4 M_4$ 4! arrangements for females, 4! arrangements for males Similarly with male first $\therefore$ number of arrangements = $2 \times 4! \times 4!$	B	1

Question 9 (1 mark)**Outcomes Assessed: E2****Targeted Performance Bands: E2**

Solution	Answer	Mark
$x^2 - 1 \geq 0$ and $x + \sqrt{x^2 - 1} > 0$ i.e $x \leq -1, x \geq 1$ $\therefore$ domain is $x \geq 1$	C	1

Question 10 (1 mark)**Outcomes Assessed: E9****Targeted Performance Bands: E4**

Solution	Answer	Mark
If we look at the number 1 it can have a value of (1000+100+10+1)=1111 with the other numbers arranged 3! ways. The value of all numbers is $3! \times (1111) \times (1+2+3+4) = 66,660$	B	1

Summary

1 C	6 A
2 A	7 C
3 D	8 B
4 A	9 C
5 C	10 B

Question 11 (15 marks)

(a) (i) (1 mark)

Outcomes assessed: E3**Targeted Performance Bands: E2**

Criteria	Marks
• Both modulus and argument correct and written in correct form	2
• Modulus or argument correct	1

$$|\sqrt{3} + i| = \sqrt{(\sqrt{3})^2 + 1^2} = 2$$

$$\arg(\sqrt{3} + i) = \frac{\pi}{6} \quad \therefore \sqrt{3} + i = 2\text{cis}\frac{\pi}{6}$$

(ii) (2 marks)

Outcomes assessed: E3**Targeted Performance Bands: E2 – E3**

Criteria	Marks
• Correct value in simplest form	3
• Some further progress towards solution	2
• Obtains $\frac{\sqrt{2}}{2}\text{cis}\left(\frac{7\pi}{12}\right)$ using De-Moivre's Theorem	1

Sample answer:

$$z = \frac{\sqrt{2}\text{cis}\left(\frac{3\pi}{4}\right)}{2\text{cis}\left(\frac{\pi}{6}\right)}$$

$$= \frac{\sqrt{2}}{2}\text{cis}\left(\frac{3\pi}{4} - \frac{\pi}{6}\right)$$

$$= \frac{\sqrt{2}}{2}\text{cis}\left(\frac{7\pi}{12}\right)$$

$$\text{Now } \frac{-1+i}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i} = \frac{1-\sqrt{3}}{4} + \frac{(1+\sqrt{3})i}{4}$$

$$\text{Equating real parts } \frac{\sqrt{2}}{2}\cos\left(\frac{7\pi}{12}\right) = \frac{1-\sqrt{3}}{4}$$

$$\therefore \cos\left(\frac{7\pi}{12}\right) = \frac{1-\sqrt{3}}{4} \times \frac{2}{\sqrt{2}} = \frac{\sqrt{2}-\sqrt{6}}{4}$$

11. (b) (i) (2 marks)

Outcomes assessed: E8

Targeted Performance Bands: E3

Criteria	Marks
• Finds correct values of A, B and C	2
• Makes some progress in partial fraction decomposition	1

Sample answer:

$$\begin{aligned}\frac{x^2 + 3x - 4}{x^2 - x - 2} &\equiv A + \frac{Bx + c}{x^2 - x - 2} \\ \frac{x^2 - x - 2 + 4x - 2}{x^2 - x - 2} &\equiv A + \frac{Bx + c}{x^2 - x - 2} \\ \frac{x^2 - x - 2}{x^2 - x - 2} + \frac{4x - 2}{x^2 - x - 2} &\equiv A + \frac{Bx + c}{x^2 - x - 2} \\ 1 + \frac{4x - 2}{x^2 - x - 2} &\equiv A + \frac{Bx + c}{x^2 - x - 2}\end{aligned}$$

Equating $A = 1$, $B = 4$ and $C = -2$

11. (b) (ii) (2 marks)

Outcomes assessed: E8

Targeted Performance Bands: E2

Criteria	Marks
• Correct integration of partial fractions from (i). No penalty for no absolute value or no +c.	2
• Some progress towards answer.	1

Sample answer:

$$\begin{aligned}\int \frac{x^2 + 3x - 4}{x^2 - x - 2} dx &= \int \left(1 + \frac{2(2x - 1)}{x^2 - x - 2} \right) dx \\ &= x + 2 \ln |x^2 - x - 2| + c\end{aligned}$$

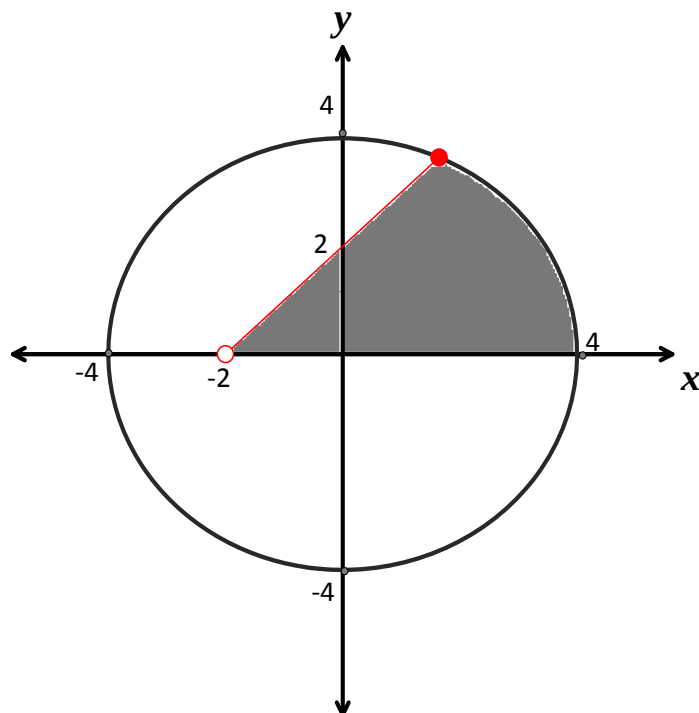
11. (c) (2 marks)Type equation here.

Outcomes assessed: E3

Targeted Performance Bands: E2

Criteria	Marks
• Correct region shown	2
• Correctly gives one region	1

Sample answer:Type equation here.



11. (d) (4 marks)

Outcomes assessed: E8

Targeted Performance Bands: E3 – E4

Criteria	Marks
• Correct answer	4
• Integrates $\tan^2 \theta \sec^2 \theta$	3
• Some progress towards evaluation of integral	2
• Correct substitution $x = 2 \sin \theta$	1

Sample answer:

Let $x = 2 \sin \theta$

$$\int_0^{\sqrt{3}} x^2 (4 - x^2)^{-\frac{5}{2}} dx = \int_0^{\frac{\pi}{3}} \frac{4 \sin^2 \theta \times 2 \cos \theta}{[4(1 - \sin^2 \theta)]^{\frac{5}{2}}} d\theta$$

$$= \int_0^{\frac{\pi}{3}} \frac{8 \sin^2 \theta \cos \theta}{32 \cos^5 \theta} d\theta$$

$$= \frac{1}{4} \int_0^{\frac{\pi}{3}} \frac{\sin^2 \theta}{32 \cos^4 \theta} d\theta$$

$$= \frac{1}{4} \int_0^{\frac{\pi}{3}} \tan^2 \theta \sec^2 \theta d\theta$$

$$= \frac{1}{12} [\tan^3 \theta]_0^{\frac{\pi}{3}}$$

$$= \frac{\sqrt{3}}{4}$$

Question 12 (15 marks)

(a) (i) (1 mark)

Outcomes assessed: E4**Targeted Performance Bands: E2**

Criteria	Marks
• Correctly shows that $(2+i)$ is a root.	1

Sample answer:

$$\begin{aligned}(2+i)^3 - 11(2+i) + 20 \\&= 8 + 12i + 6i^2 + i^3 - 22 - 11i + 20 \\&= 8 + 12i - 6 - i - 22 - 11i + 20 \\&= 0 \\&\therefore (2+i) \text{ is a root of } x^3 - 11x + 20 = 0\end{aligned}$$

(a) (ii) (2 marks)

Outcomes assessed: E4**Targeted Performance Bands: E2 – E3**

Criteria	Marks
• Correctly solves the equation	2
• Correctly identifies that $2-i$ is also a root.	1

Sample answer:

$$\begin{aligned}(2-i) \text{ is a root, conjugate pairs.} \\ \text{Now } (2+i) + (2-i) + \alpha = 0 \\ \text{So } \alpha = -4 \\ \therefore \text{Solutions are } 2+i, 2-i \text{ and } -4\end{aligned}$$

12. (b) (i) (2 marks)

Outcomes assessed: E3

Targeted Performance Bands: E2

Criteria	Marks
• Correctly derives the equation of the normal	2
• Finds gradient of normal	1

Sample answer:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

differentiate w.r.t.x

$$\frac{2x}{a^2} + \frac{2y}{b^2} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

At $P(a \cos \theta, b \sin \theta)$

$$\frac{dy}{dx} = -\frac{b \cos \theta}{a \sin \theta}$$

$$M_N = \frac{a \sin \theta}{b \cos \theta}$$

equation of normal

$$y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

$$\frac{by}{\sin \theta} - b^2 = \frac{ax}{\cos \theta} - a^2$$

$$by \cos \theta - b^2 = ax \sec \theta - a^2$$

$$\therefore ax \sec \theta - by \cos \theta = a^2 - b^2$$

12. (b) (ii) (2 marks)

Outcomes assessed: E3

Targeted Performance Bands: E3

Criteria	Marks
Correctly establishes the result.	2
Some progress towards establishing the result (e.g. finds OG and attempts to use $b^2 = a^2(1 - e^2)$).	1

Sample answer:

Put $y = 0$ into equation of normal

$$ax \sec \theta = a^2 - b^2$$

$$x = \frac{a^2 - b^2}{a \sec \theta} \rightarrow OG = \frac{a^2 - b^2}{a \sec \theta}$$

Using $b^2 = a^2(1 - e^2)$

$$OG = \frac{a^2 - a^2(1 - e^2)}{a \sec \theta} = \frac{ae^2}{\sec \theta} = ae^2 \cos \theta$$

Now

$$ON = a \cos \theta$$

$$\therefore OG : ON = ae^2 \cos \theta : a \cos \theta = e^2 : 1$$

12. (b) (iii) (1 mark)

Outcomes assessed: E3

Targeted Performance Bands: E4

Criteria	Marks
Correctly establishes the ratio	1

Sample answer:

Using similar triangles

$$OG : GN = e^2 : 1 - e^2$$

$$\therefore \Delta PNG : PN'G' = (1 - e^2)^2 : 1$$

12. (c) (3 marks)

Outcomes assessed: E4

Targeted Performance Bands: E4

Criteria	Marks
• Correctly solves the equation	3
• Correctly factorises $P(x)$	2
• Shows that $x = -2$ is a root of multiplicity 3.	1

Sample answer:

$$P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$$

$$P'(x) = 8x^3 + 27x^2 + 12x - 20$$

$$P''(x) = 24x^2 + 54x + 12$$

if $x = \alpha$ is a root of multiplicity 3 of $P(x) = 0$

then it is a root of $P''(x) = 0$.

$$6(4x^2 + 9x + 2) = 0$$

$$6(4x + 1)(x + 2) = 0$$

$$\therefore x = -2 \text{ or } -\frac{1}{4}$$

$$\text{now } P(-2) = 32 - 72 + 24 + 40 - 24 = 0$$

then $x = -2$ is a root with multiplicity 3.

$$\text{so } P(x) = (x + 2)^3(ax + b)$$

from inspection $a = 2$

substitute $x = 0$

$$-24 = 2^3 \times b \rightarrow b = -3$$

$$\therefore \text{solution of } P(x) = 0 \text{ is } x = -2 \text{ or } \frac{3}{2}$$

12. (d) (4 marks)

Outcomes assessed: E7

Targeted Performance Bands: E3

Criteria	Marks
• Correct answer	4
• Some progress towards evaluation of volume	3
• Some progress towards evaluation of cross-section	2
• Correctly finds point of intersection	1

Sample answer:

Solve simultaneously $x = y^2$ and $x = 3 - 2y^2$

$$y^2 = 3 - 2y^2$$

$$3y^2 = 3$$

$$\therefore y = \pm 1$$

$$x = 1$$

from $x = 0$ to $x = 1$, the cross section has area

$$A_1 = (2y)^2 = 4y^2 = 4x$$

the volume from $x = 0$ to $x = 1$ is:

$$V_1 = \int_0^1 4x \, dx = 2 \left[x^2 \right]_0^1 = 2$$

from $x = 1$ to $x = 3$, the cross section has area

$$A_2 = (2y)^2 = 4y^2 = 2(3 - x)$$

the volume from $x = 1$ to $x = 3$ is:

$$\begin{aligned} V_2 &= 2 \int_1^3 (3 - x) \, dx = 2 \left[3x - \frac{x^2}{2} \right]_1^3 \\ &= 2 \left[(9 - 4.5) - (3 - 0.5) \right] = 4 \end{aligned}$$

$\therefore$ the volume of the solid is $2 + 4 = 6u^3$

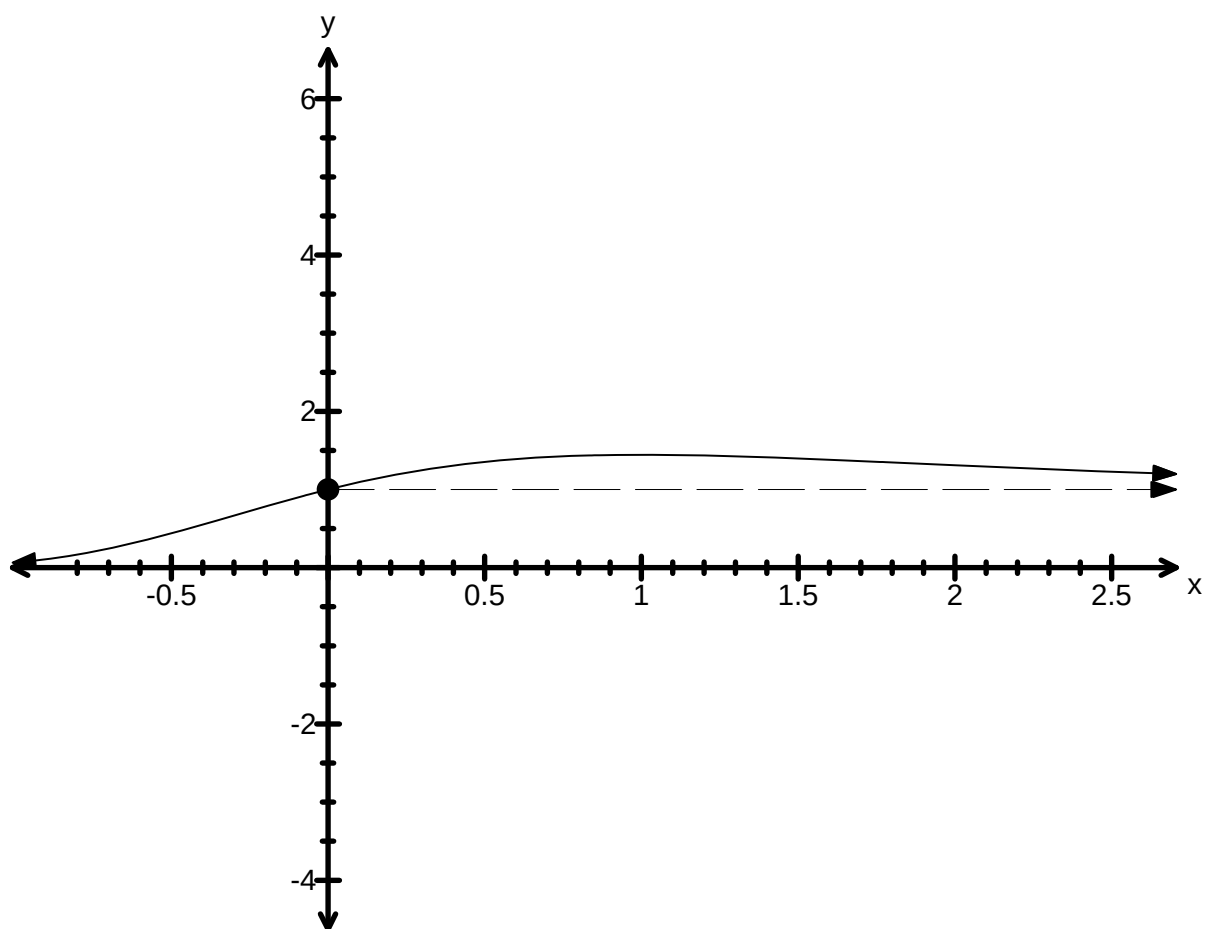
Question 13 (15 marks)

13. (a) (i) (2 marks)

Outcomes assessed: E6**Targeted Performance Bands: E2 - E3**

Criteria	Marks
• Sketches correct curve	2
• Sketches curve for $x \geq 0$ or for $x < 0$ showing asymptote	1

Sample answer:



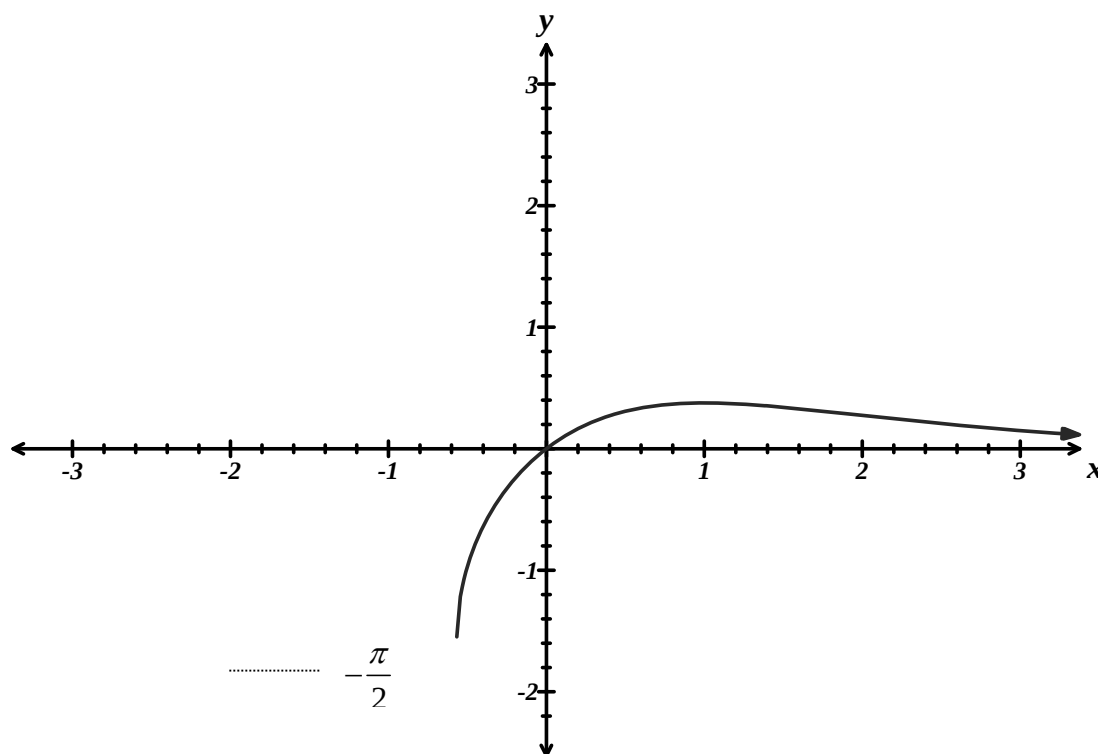
13. (a) (ii) (3 marks)

Outcomes assessed: E6

Targeted Performance Bands: E3 – E4

Criteria	Marks
• Sketches correct curve indicating $y = -\frac{\pi}{2}$.	3
• Sketches curve for $x < 0$	2
• Sketches curve for $x \geq 0$ showing asymptote	1

Sample answer:



13. (b) (2 marks)

Targeted Performance Bands: E3

Criteria	Marks
• Correct answer	2
• Some progress towards answer.	1

Sample answer:

The chairperson can be chosen in ${}^{11}C_1$ ways

The other 3 members can be chosen in ${}^{10}C_3$ ways

$\therefore$ number of committees = ${}^{11}C_1 \times {}^{10}C_3 = 1320$

13. (c) (3 marks)

Targeted Performance Bands: E3

Criteria	Marks
• Correctly establishes result.	3
• Correctly applies completing of the square and recognises to use $\sin^{-1}$.	2
• Attempts to use completing of the square	1

Sample answer:

$$\begin{aligned}\int \frac{1}{\sqrt{3-(x^2-2x)}} dx &= \int \frac{1}{\sqrt{4-(x-1)^2}} dx \\ &= \sin^{-1}\left(\frac{x-1}{2}\right) + c\end{aligned}$$

13. (d) (2 marks)

Targeted Performance Bands: E3

Criteria	Marks
• Correctly determines the integral.	2
• Attempts to use integration by parts	1

Sample answer:

$$\begin{aligned}\int \tan^{-1} x \, dx &= \int 1 \cdot \tan^{-1} x \, dx \\ &= x \tan^{-1} x - \int \frac{x}{1+x^2} dx \\ &= x \tan^{-1} x - \frac{1}{2} \log_e(1+x^2) + c\end{aligned}$$

13. (e) (3 marks)

Outcomes assessed:

Targeted Performance Bands: E3 – E4

Criteria	Marks
• Correct solution	3
• Finds the stationary point and attempts to test its nature	2
• Some progress towards answer.	1

Sample answer:

$$\text{Let } f(x) = \cos x - 1 + \frac{x^2}{2} > 0, \text{ if } x \neq 0.$$

$$f'(x) = -\sin x + x$$

$$f''(x) = -\cos x + 1 \geq 0 \text{ for all } x, x \neq 0$$

Stationary point occurs when $f'(x) = 0$

ie when $\sin x = x$

$$x = 0$$

When $x = 0, y = 0$

$\therefore$ Stationary point of inflection at (0,0)

$f''(0) = 0$, $\therefore$ possible point of inflection at (0,0)

Test $f''(x)$ about $x = 0$

x	$x < 0$	$x = 0$	$x > 0$
$f''(x)$	-	0	+

Therefore minimum turning point

$\therefore$ there is an absolute minimum at (0,0)

$\therefore f(0) < f(x)$ for all $x \neq 0$

$$\therefore 0 < \cos x - 1 + \frac{x^2}{2}$$

Question 14 (15 marks)

(a) (i) (1 mark)

Outcomes assessed: E9**Targeted Performance Bands: E2**

Criteria	Marks
• Correctly shows that $y = f(x)$ is an even function.	1

Sample answer:

$$f(x) = \frac{3}{14}x^{\frac{2}{3}}\left(x^4 - \frac{7}{2}x^2 + 7\right)$$

$$f(-x) = \frac{3}{14}(-x)^{\frac{2}{3}}\left((-x)^4 - \frac{7}{2}(-x)^2 + 7\right)$$

$$= \frac{3}{14}x^{\frac{2}{3}}\left(x^4 - \frac{7}{2}x^2 + 7\right)$$

$$= f(x) \quad \therefore y = f(x) \text{ is an even function.}$$

14. (a) (ii) (2 marks)

Outcomes assessed: E6

Targeted Performance Bands: E3

Criteria	Marks
• Correctly determines the nature of the stationary points.	2
• Uses appropriate method to determine the nature of the stationary point.	1

Sample answer:

$$f(x) = \frac{3}{14}x^{\frac{2}{3}}\left(x^4 - \frac{7}{2}x^2 + 7\right)$$

$$= \frac{3}{14}x^{\frac{14}{3}} - \frac{3}{4}x^{\frac{8}{3}} + \frac{3}{2}x^{\frac{2}{3}}$$

$$f'(x) = x^{\frac{11}{3}} - 2x^{\frac{5}{3}} + x^{-\frac{1}{3}}$$

$$f''(x) = \frac{11}{3}x^{\frac{8}{3}} - \frac{10}{3}x^{\frac{2}{3}} - \frac{1}{3}x^{-\frac{4}{3}}$$

since Stationary points are $\left(1, \frac{27}{28}\right)$ and $\left(-1, \frac{27}{28}\right)$

$$f''(1) = 0$$

x	0.9	1	1.1
$f''(x)$	-	0	+

$\therefore$ Both points are Horizontal points of inflexion since $y = f(x)$ is an even function.

14. (a) (iii) (1 mark)

Outcomes assessed: E6

Targeted Performance Bands: E3

Criteria	Marks
• Correctly answer.	1

Sample answer:

$f'(x)$ is undefined at the origin.

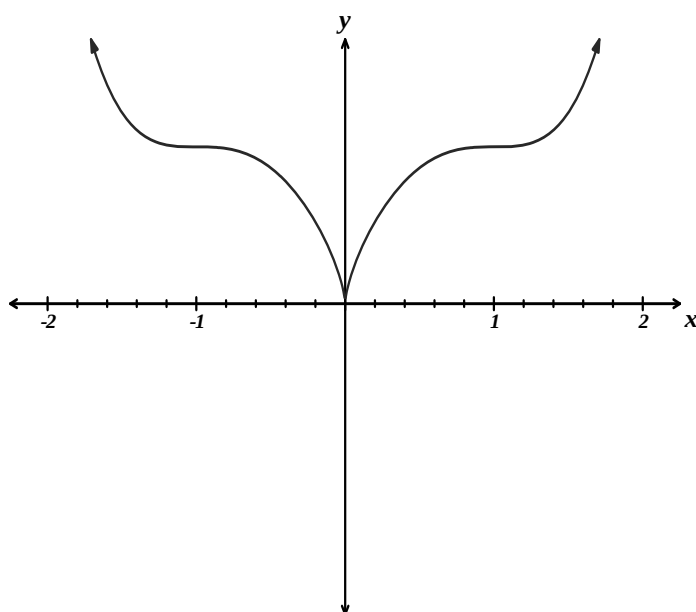
(a) (iv) (2 marks)

Outcomes assessed: E6

Targeted Performance Bands: E3

Criteria	Marks
• Correctly draws $y = f(x)$ showing important features.	2
• Draws part of $y = f(x)$ correctly.	1

Sample answer:



14. (b) (i) (3 marks)

Outcomes assessed: E8

Targeted Performance Bands: E4

Criteria	Marks
• Correctly establishes result.	3
• Also correctly establishes limits	2
• Correct substitution of $t = \tan \frac{\theta}{2}$ and use of $\frac{d\theta}{dt} = \frac{2}{1+t^2}$ in integral	1

Sample answer:

If $\tan \frac{\theta}{2} = t$, then as $\theta \rightarrow \pi, t \rightarrow \infty$

Hence

$$\begin{aligned} I &= \lim_{h \rightarrow \infty} \int_0^h \frac{1+t^2}{a(1+t^2)+b(1-t^2)} \times \frac{2}{1+t^2} dt \\ &= \lim_{h \rightarrow \infty} \int_0^h \frac{2dt}{a+b+t^2(a-b)} \\ &= \lim_{h \rightarrow \infty} \int_0^h \frac{2dt}{(a-b) \left[t^2 + \frac{a+b}{a-b} \right]} \\ &= \lim_{h \rightarrow \infty} \int_0^h \frac{2dt}{(a-b)(t^2 + c^2)} \quad \text{where } c^2 = \frac{a+b}{a-b} \end{aligned}$$

14. (b) (ii) (2 marks)

Outcomes assessed: E8

Targeted Performance Bands: E4

Criteria	Marks
• Correctly establishes the result.	2
• correctly integrates $\int_0^\infty \frac{dt}{(t^2 + c^2)}$	1

Sample answer:

$$\begin{aligned}
 2 \int_0^\infty \frac{dt}{(a-b)(t^2 + c^2)} &= \frac{2}{a-b} \lim_{h \rightarrow \infty} \int_0^h \frac{dt}{(t^2 + c^2)} \\
 &= \frac{2}{c(a-b)} \lim_{h \rightarrow \infty} \left[\tan^{-1} \frac{t}{c} \right]_0^h, \text{ now as } h \rightarrow \infty, \tan^{-1} \frac{t}{c} \rightarrow \frac{\pi}{2} \\
 &= \frac{2}{c(a-b)} \left[\frac{\pi}{2} - 0 \right] \\
 &= \frac{\pi}{c(a-b)} \\
 &= \frac{\pi}{\left[\sqrt{\frac{a+b}{a-b}} \right] (a-b)}, c^2 = \frac{a+b}{a-b} \\
 &= \frac{\pi}{\sqrt{(a^2 - b^2)}}
 \end{aligned}$$

14. (c) (3 marks)

Outcomes assessed: E9

Targeted Performance Bands: E3

Criteria	Marks
• Correct answer.	4
• Uses integration by parts	3
• Correct use of the shell method	2
• Attempt to use the shell method	1

Sample answer:

By shell Method

$$V = 2\pi \int_0^\pi x \sin x dx$$

$$\text{let } u = x$$

$$v' = \sin x$$

$$u' = 1$$

$$v = -\cos x$$

$$I = uv - \int vu'$$

$$= 2\pi[-x\cos x]_0^\pi - \int_0^\pi -\cos x \, dx$$

$$= 2\pi[\pi + (\sin x)_0^\pi]$$

$$= 2\pi^2$$

Question 15 (15 marks)

(a) (i) (1 mark)

Outcomes assessed: E9

Targeted Performance Bands: E3

Criteria	Marks
Correctly establishes the result.	1

Sample answer:

$$(3 + \sqrt{5}) + (3 - \sqrt{5}) = 6 = 2^1 \times 3$$

$$\therefore \text{true } n = 1$$

$$(3 + \sqrt{5})^2 + (3 - \sqrt{5})^2 = 9 + 6\sqrt{5} + 5 + 9 - 6\sqrt{5} + 5 = 28 = 2^2 \times 7$$

$$\therefore \text{true } n = 2$$

15. (a) (ii) (3 marks)

Outcomes assessed: E2, E9

Targeted Performance Bands: E3-E4

Criteria	Marks
• Correctly proof by mathematical induction.	3
• Makes two correct assumptions and attempts to prove for $n = k + 1$ or correctly uses one assumption	2
• Attempts to use one assumption	1

Sample answer:

Step1: From (i) true for $n = 1$ and $n = 2$

Step 2: Let $n = k$ and $n = k - 1$ be values for which the statement is true

$$\text{i.e } (3 + \sqrt{5})^k + (3 - \sqrt{5})^k = 2^k \times M$$

$$\text{and } (3 + \sqrt{5})^{k-1} + (3 - \sqrt{5})^{k-1} = 2^{k-1} \times N$$

M and N are integers.

Step 3: Prove true for $n = k + 1$

$$\text{Now } (3 + \sqrt{5})^{k+1} + (3 - \sqrt{5})^{k+1} = p^{k+1} + q^{k+1}, p = 3 + \sqrt{5} \text{ and } q = 3 - \sqrt{5}$$

$$= p^k p + q^k q$$

$$= [2^k \times M - q^k] p + [2^k \times M - p^k] q \text{ from step 2}$$

$$= 2^k \times M (p + q) - pq (q^{k-1} + p^{k-1})$$

$$= 2^k \times 6M - 4 [2^{k-1} \times N] \text{ from step 2}$$

$$= 2^{k+1} \times 3M - 2^{k+1} \times N$$

$$= 2^{k+1} [3M - N]$$

$\therefore$ True by mathematical induction

(b) (3 marks)

Outcomes assessed: E3

Targeted Performance Bands: E4

Criteria	Marks
• Finds correct locus with restriction.	3
• Substitutes $a^2 + b^2 = 1$	2
• Finds expression of z in terms of a and b	1

Sample answer:

let $z = x + iy$ and $w = a + ib$

$$\begin{aligned}\therefore x + iy &= \frac{1+a+ib}{1-a-ib} \times \frac{1-a+ib}{1-a+ib} \\ &= \frac{1-a+ib+a-a^2+iab+ib-iab-b^2}{(1-a)^2+b^2} \\ &= \frac{1-a^2-b^2+2ib}{(1-a)^2+b^2} \\ &= \frac{2ib}{(1-a)^2+b^2}, \text{ since } |w|=1 \rightarrow a^2+b^2=1\end{aligned}$$

$\therefore x = 0$

z is purely imaginary

$\therefore$ locus of z is the y – axis, excluding the origin.

15. (c) (i) (1 mark)

Outcomes assessed: E5

Targeted Performance Bands: E3

Criteria	Marks
• Correctly establishes the result.	1

Sample answer:

$$\downarrow mg \quad \uparrow R = \frac{7v^2}{10}$$

$$F = m\ddot{x}$$

$$m\ddot{x} = mg - R$$

$$= mg - \frac{7v^2}{10}$$

$$\ddot{x} = g - \frac{7v^2}{10m}, \quad m = 7 \text{ and } g = 10$$

$$\therefore \ddot{x} = 10 - \frac{v^2}{10}$$

15. (c) (ii) (1 mark)

Outcomes assessed: E2

Targeted Performance Bands: E3

Criteria	Marks
• Correctly establishes the result.	1

Sample answer:

$$\ddot{x} = 10 - \frac{v^2}{10}$$

$$\ddot{x} \rightarrow 0$$

$$v^2 \rightarrow 100$$

$$\therefore \text{Terminal velocity is } v = 10 \text{ m / s}$$

15. (c) (iii) (3 marks)

Outcomes assessed: E5

Targeted Performance Bands: E3-E4

Criteria	Marks
• Correct answer	3
• Correct integration.	2
• Correctly uses $\ddot{x} = v \frac{dv}{dx}$ plus some further progress.	1

Sample answer:

$$\ddot{x} = 10 - \frac{v^2}{10}$$

$$v \frac{dv}{dx} = 10 - \frac{v^2}{10}$$

$$\frac{dv}{dx} = \frac{10}{v} - \frac{v}{10} = \frac{100 - v^2}{10v}$$

$$\frac{dx}{dv} = \frac{10v}{100 - v^2}$$

$$x = \int \frac{10v}{100 - v^2} dv$$

$$\therefore x = -5 \ln(100 - v^2) + c$$

$$x = 0, v = 0$$

$$c = 5 \ln(100)$$

$$\therefore x = 5 \ln\left(\frac{100}{100 - v^2}\right)$$

$$e^{-x/5} = \frac{100 - v^2}{100}$$

$$\therefore v^2 = 100\left(1 - e^{-x/5}\right)$$

15. (c) (iv) (3 marks)

Outcomes assessed: E5

Targeted Performance Bands: E3-E4

Criteria	Marks
• Correct answer.	3
• Integrates with correct limits.	2
• Correctly recognises $\ddot{x} = dV/dt$ plus some further progress.	1

Sample answer:

$$\ddot{X} = -10 - \frac{V^2}{10}$$

$$\frac{dV}{dt} = -10 - \frac{V^2}{10}$$

$$\frac{dt}{dV} = \frac{-10}{100 + V^2}, \text{ at maximum height } V = 0$$

$$\therefore t = -10 \int_{10/\sqrt{3}}^0 \frac{dV}{100 + V^2} = 10 \int_0^{10/\sqrt{3}} \frac{1}{100 + V^2} dV$$

$$= \left[\tan^{-1} \frac{V}{10} \right]_0^{10/\sqrt{3}}$$

$$= \left[\left(\frac{\pi}{6} \right) - 0 \right]$$

$$\therefore t = \frac{\pi}{6} \text{ seconds}$$

Question 16 (15 marks)

(a) (i) (3 marks)

Outcomes assessed: E3

Targeted Performance Bands: E3-E4

Criteria	Marks
• Establish Correct result	3
• Expanding using Binomial Theorem	2
• Correct use of De Moivre's Theorem	1

Sample answer:

By De Moivre's Theorem

$$\begin{aligned}\cos 7\theta + i \sin 7\theta &= (\cos \theta + i \sin \theta)^7 \\ &= \cos^7 \theta (1 + it)^7, \text{ where } t = \tan \theta\end{aligned}$$

Expanding by the Binomial Theorem

$$= \cos^7 \theta (1 + {}^7C_1(it) - {}^7C_2t^2 - {}^7C_3(it^3) + {}^7C_4t^4 + {}^7C_5(it^5) - {}^7C_6t^6 - it^7)$$

equating real and imaginary parts

$$\cos 7\theta = \cos^7 \theta (1 - {}^7C_2t^2 + {}^7C_4t^4 - {}^7C_6t^6) \dots\dots\dots(1)$$

$$\sin 7\theta = \cos^7 \theta ({}^7C_1t - {}^7C_3t^3 + {}^7C_5t^5 - t^7) \dots\dots\dots(2)$$

$$(2) \div (1)$$

Hence

$$\tan 7\theta = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6}$$

16. (a) (ii) (3 marks)

Outcomes assessed: E3

Targeted Performance Bands: E4

Criteria	Marks
• Establish Correct result	3
• Establish equation $x^3 - 21x^2 + 35x - 7 = 0$	2

• Find roots of $7t - 35t^3 + 21t^5 - t^7 = 0$	1
--	----------

Sample answer:

Consider the equation

$$\tan 7\theta = 0 \rightarrow 7\theta = n\pi, \text{ for integer } n.$$

If $\tan 7\theta = 0$, then

$$7t - 35t^3 + 21t^5 - t^7 = 0, \text{ when}$$

$$t = 0, \tan\left(\frac{\pi}{7}\right), \tan\left(\frac{2\pi}{7}\right), \tan\left(\frac{3\pi}{7}\right), \tan\left(\frac{4\pi}{7}\right), \tan\left(\frac{5\pi}{7}\right) \text{ and } \tan\left(\frac{6\pi}{7}\right)$$

Now divide by $-t$ we get ($t \neq 0$)

$$t^6 - 21t^4 + 35t^2 - 7 = 0$$

Let $t = x^2$

$$x^3 - 21x^2 + 35x - 7 = 0$$

$$\therefore x = \tan^2\left(\frac{\pi}{7}\right), \tan^2\left(\frac{2\pi}{7}\right) \text{ and } \tan^2\left(\frac{3\pi}{7}\right) \text{ are solutions of the equation.}$$

16. (a) (iii) (2 marks)

Outcomes assessed: E4

Targeted Performance Bands: E3

Criteria	Marks
• Correctly establishes the result.	2
• Some progress towards the result.	1

Sample answer:

Let p, q and r be the roots of the equation

Now the required sum is

$$\begin{aligned}
 (1+p)^2 + (1+q)^2 + (1+r)^2 &= 3 + 2(p+q+r) + p^2 + q^2 + r^2 \\
 &= 3 + 2(p+q+r) + (p+q+r)^2 - 2(pq+rq+pq) \\
 &= 3 + 42 + 441 - 70 \\
 &= 416
 \end{aligned}$$

16. (b) (i) (2 marks)

Outcomes assessed: E9

Targeted Performance Bands: E3

Criteria	Marks
• Correctly establishes the result.	2
• Some progress towards the result.	1

Sample answer:

$$\begin{aligned}
S_n &= 1 + x + x^2 + \dots + x^n \\
xS_n &= x + x^2 + x^3 + \dots + x^{n+1} \\
S_n(1-x) &= 1 - x^{n+1} \\
\therefore S_n &= \frac{1 - x^{n+1}}{1 - x} \\
\text{i.e } 1 + x + x^2 + \dots + x^{n-1} + x^n &= \frac{1 - x^{n+1}}{1 - x} \\
1 + x + x^2 + \dots + x^{n-1} &= \frac{1 - x^{n+1}}{1 - x} - x^n \\
&= \frac{1 - x^{n+1} - x^n + x^{n+1}}{1 - x} \\
&= \frac{1 - x^n}{1 - x} = \frac{1}{1 - x} - \frac{x^n}{1 - x} \\
\text{thus } \frac{1}{1 - x} &= 1 + x + x^2 + \dots + x^{n-1} + \frac{x^n}{1 - x}
\end{aligned}$$

16. (b) (ii) (3 marks)

Outcomes assessed: E8

Targeted Performance Bands: E4

Criteria	Marks
• Derives correct result	3
• Integrates with correct limits	2
• Attempts to integrate both sides after making substitution	1

Sample answer:

Let $x = -t$, then

$$\frac{1}{1+t} = 1 + (-t) + (-t)^2 + \dots + (-t)^{n-1} + \frac{(-t)^n}{1+t}$$

Integrating both sides

$$\int_0^x \frac{1}{1+t} dt = \int_0^x \left(1 - t + t^2 - \dots + (-t)^{n-1} + \frac{(-t)^n}{1+t} \right) dt$$

$$\left[\log_e(1+t) \right]_0^x = \left[t - \frac{t^2}{2} + \frac{t^3}{3} - \dots + (-1)^{n-1} \frac{t^n}{n} \right]_0^x + (-1)^n \int_0^x \frac{t^n}{1+t} dt$$

$$\therefore \log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^{n-1} \frac{x^n}{n} + T_n(x)$$

$$\text{where } T_n(x) = (-1)^n \int_0^x \frac{t^n}{1+t} dt$$

16. (b) (iii) (1 marks)

Outcomes assessed: E8

Targeted Performance Bands: E4

Criteria	Marks
• Correctly establishes the result.	1

Sample answer:

$$|T_n(x)| = \int_0^x \frac{t^n}{1+t} dt \leq \int_0^x t^n dt = \frac{x^{n+1}}{n+1} \rightarrow 0 \text{ as } n \rightarrow \infty \text{ and } 0 \leq x \leq 1$$

16. (b) (iv) (1 marks)

Outcomes assessed: E9

Targeted Performance Bands: E3

Criteria	Marks
• Correct answer	1

Sample answer:

$x = 1$ in (ii)

$$\text{Let } \therefore \log_e 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$$